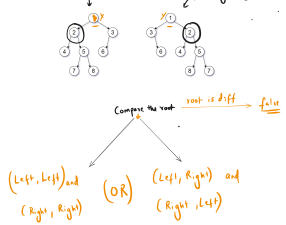
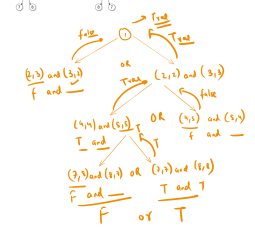


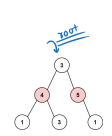
Binary Tree
Leetcode 951



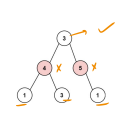
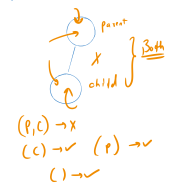
$isSameTree(root1, root2) =$
 returns whether root1 & root2 are the same or not?
 (root1.left, root2.left) and (root1.right, root2.right)
 Base case: root1 == null and root2 == null -> return true



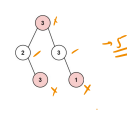
Dp 0-1 Trees -> (Repeated Subproblem)
 (Distinct)



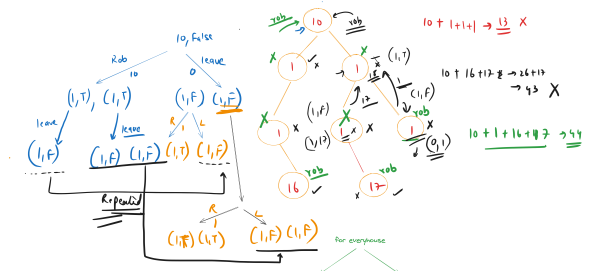
Houses are arranged in form of Binary Trees



$$3 + 1 + 3 + 1 = 8 < 9$$



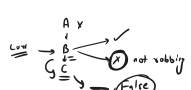
$f(\text{root}, \text{False}) = \text{ans}$



Either leave the house no matter what
 Doesn't matter if parent was rob or not - we leave it
 If the parent is not rob then rob this house

Dp has to be 0-1

$isParentRobbed = \text{False}$
 $leave = \text{robLeft, False} + \text{robRight, False}$
 $rob = \text{not rob} + \text{robLeft, True} + \text{robRight, True}$
 $\rightarrow \text{max(leave, take)}$
 else -> parent was robbed
 return $\text{robLeft, False} + \text{robRight, False}$



Base case: root == null -> return 0